NAMAN SAINI

AI and Data Professional | Python, SQL, Machine Learning and Analytics

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UK Student visa holder; planning to apply for the Graduate visa, subject to eligibility and approval.

PROFILE

MSc Artificial Intelligence student at the University of Bath, with final results pending. Builds Python applications and analytical pipelines that turn retail, fraud and security data into testable outputs and practical recommendations. Portfolio experience spans transformer fine-tuning, retrieval-assisted investigation, relational databases and interactive reporting; teaching experience supports clear technical communication.

TECHNICAL SKILLS

Programming and data: Python, pandas, NumPy, SQL, MySQL, relational modelling, ETL, data-quality checks

AI and ML: scikit-learn, XGBoost, PyTorch, Hugging Face Transformers, TF-IDF, MiniLM fine-tuning

Applications and reporting: FastAPI, Pydantic, MCP tools, Streamlit, Plotly, KPI reporting, forecasting

Methods and tools: Class-imbalance evaluation, threshold analysis, RFM segmentation, Git, Jupyter, testing

EXPERIENCE

Graduate Teaching Assistant | University of Bath | Nov 2025–May 2026

- Supported Python programming labs, debugging and data manipulation exercises; explained technical concepts clearly and guided systematic problem-solving.

UPSC Civil Services Examination preparation | India | 2021–2025

AI AND MACHINE LEARNING PROJECTS

Cybersecurity SOC Copilot | GitHub

Python, FastAPI, Pydantic, Streamlit, MCP | Synthetic-data functional demo

- Built a deterministic investigation workflow to triage alerts, correlate evidence, retrieve cited playbooks with TF-IDF and produce explainable risk scores.
- Added scoped MCP tools, an analyst approval audit trail and JSON/PDF incident exports; included automated checks and an evaluation harness with three labelled synthetic cases.

NLP Text Classification and Transformer Fine-tuning | GitHub

Python, PyTorch, Hugging Face Transformers, scikit-learn, Streamlit

- Built a six-class text-tone classifier and compared TF-IDF, frozen MiniLM embeddings and end-to-end MiniLM fine-tuning using class-weighted loss.
- Recorded macro-F1 of 0.915 for fine-tuned MiniLM versus 0.866 for TF-IDF in the project evaluation; retained TF-IDF as the lightweight app default and documented label-mapping limitations.

Financial Fraud Detection Pipeline | GitHub

Python, scikit-learn, XGBoost, Streamlit | Offline portfolio project

- Compared four classifiers on 283,726 deduplicated transactions, using precision-recall metrics to evaluate severe class imbalance.
- Built a transaction-scoring interface and threshold simulator to compare missed fraud and false alarms under illustrative cost assumptions; documented findings in technical and business reports.

ANALYTICS AND DATA ENGINEERING PROJECTS

E-commerce Sales Analytics | GitHub

Python, pandas, Streamlit, Plotly, statsmodels

- Analysed 9,993 cleaned Superstore order lines; identified a $17.7K loss in the Tables sub-category and examined the relationship between discounting and profitability.
- Built a KPI dashboard, a 12-month Holt-Winters forecast and RFM segmentation; identified that the top 20% of customers contributed 48% of revenue.

Airline Delay and Operational Performance Analytics | GitHub

Python, pandas, Streamlit, Plotly

- Analysed approximately 5.82 million US domestic flights from 2015; reported 81.4% on-time performance and segmented delay patterns by airline, airport and time.
- Built an executive dashboard using precomputed aggregates to explore punctuality, cancellations and delay causes without loading the full raw dataset.

Cybersecurity Incident Analytics | GitHub

Python, pandas, Streamlit, Plotly | Synthetic-data portfolio project

- Analysed 18,000 synthetic incident records and built a four-tab dashboard covering detection time, response time, SLA breaches and financial-impact KPIs, with documented data limitations.

Relational Database Design — Airline Reservation System

SQL, MySQL, Java

- Designed a normalised schema across five or more interconnected tables, with booking and seat-availability queries and transaction management for concurrent bookings.

Data Cleaning and ETL Pipeline

Python, pandas, Jupyter Notebook

- Processed approximately 50,000 customer-sales records; removed approximately 1,200 duplicates and reduced null entries by approximately 35%, documenting imputation, type conversion and outlier checks.

EDUCATION AND CERTIFICATIONS

MSc Artificial Intelligence | University of Bath | 2026

Final results pending. Dissertation: comparative analysis of machine learning and deep learning for network intrusion detection, with feature engineering and class imbalance handling.

BTech Computer Science Engineering | Amity University, Noida | 2017–2021

Oracle AI Foundation Associate | Oracle Generative AI Professional